Q1 A coil is wound with 300 turns on a square former having sides 50 mm in length. The coil rotates at 2000 r/min in a uniform magnetic field of density 0.8 T.

i. Calculate the frequency of the e.m.f.
ii. Calculate the period of the e.m.f.
iii. Calculate the maximum value of the e.m.f. generated in the coil.
iv. Write the mathematical expression of the instantaneous e.m.f. generated.

Q2 For a waveform $x(t)$ shown in Figure Q2, determine,

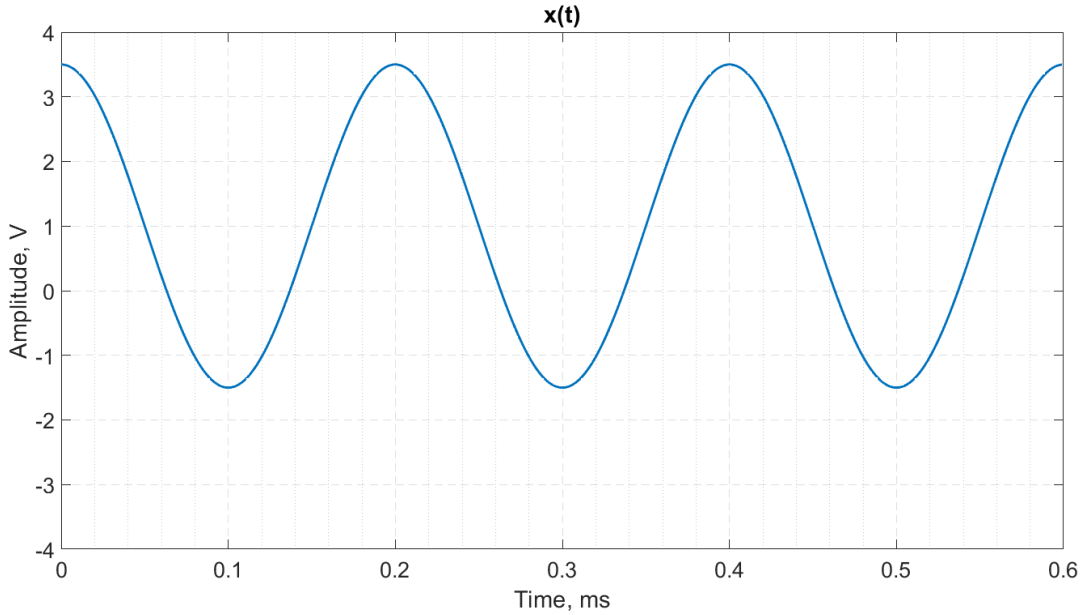


Figure Q2

i. The period of the waveform
ii. The frequency of the waveform
iii. The peak voltage of the waveform
iv. The peak-to-peak voltage of the waveform
v. The peak amplitude of the waveform
vi. The average voltage of the waveform
vii. The r.m.s. voltage of the waveform
viii. The form factor of the waveform
ix. The crest factor of the waveform
x. The mathematical expression of the waveform

Q3 The mathematical expression of an a.c. waveform is given as,

$$x(t) = 8\cos\left(377t + \frac{\pi}{6}\right)$$

Express the waveform in,

i. Phasor/polar form, $M\angle\phi$
ii. Complex/rectangular form, $A + jB$
iii. Exponential form, $Me^{j\phi}$

Q4 Given these sinusoids,

$$x(t) = 6\cos(50t - 40^{\circ})$$

$$y(t) = -4\cos(50t + 50^{\circ})$$

Calculate and write the sinusoid expression for the phasor below.

i. $a(t) = y(t) + x(t)$
ii. $b(t) = y(t) - x(t)$
iii. $c(t) = y(t) \times x(t)$

Draw all the sinusoids on a phasor diagram.